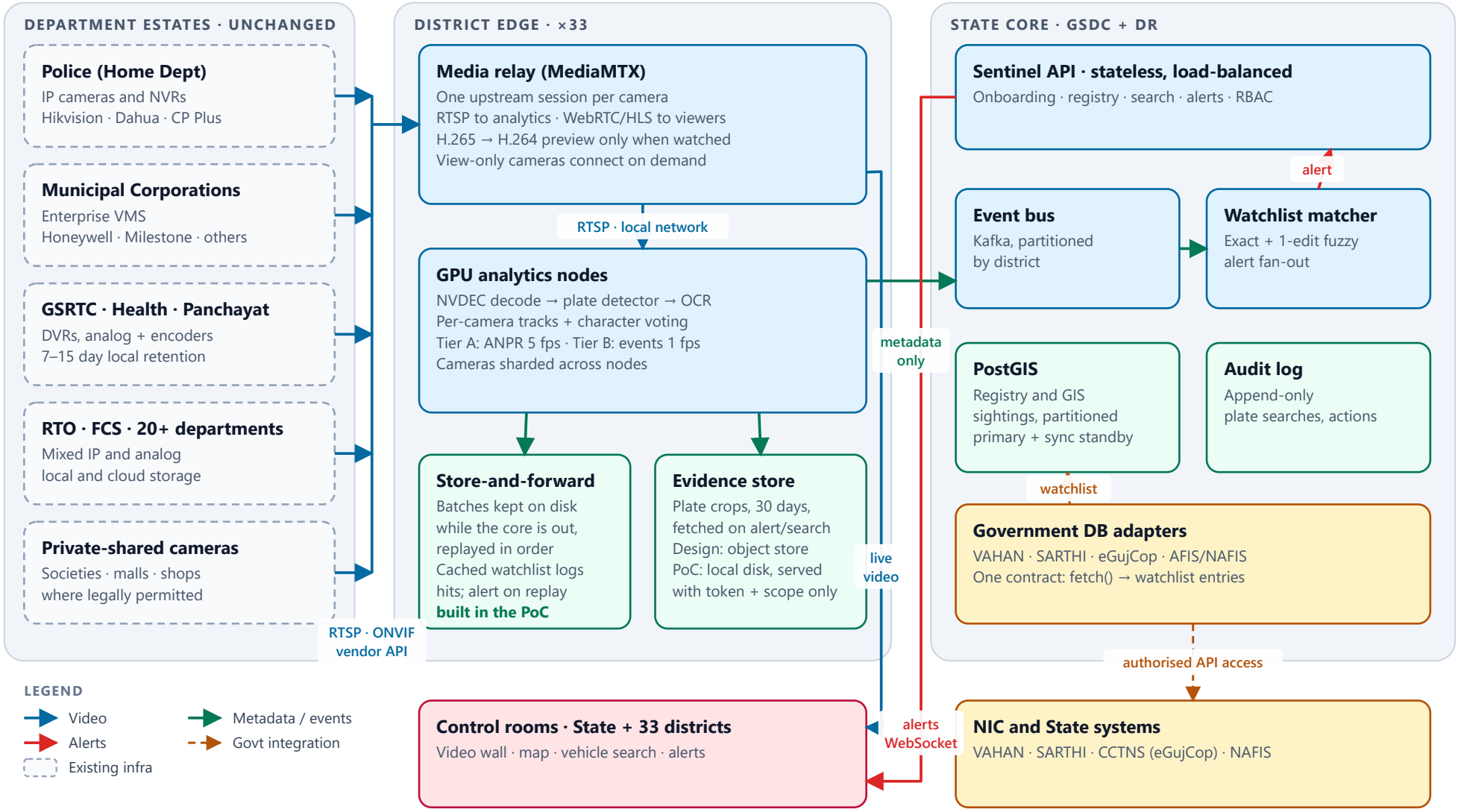
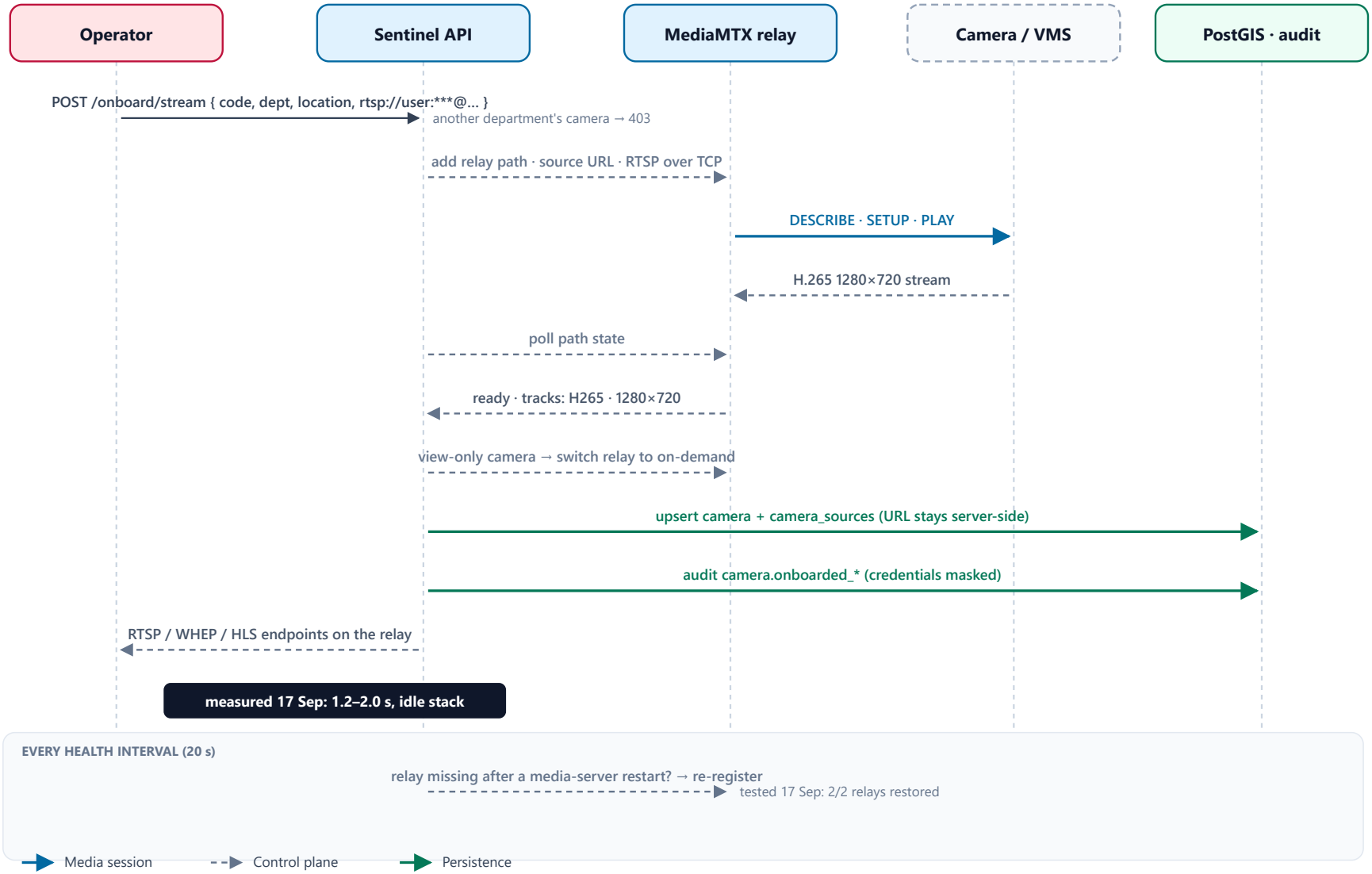
Statewide architecture (target design)

Department estates are untouched; district edge nodes relay and analyse video; only metadata reaches the state core, and a plate crop when an alert or search asks for it. Store-and-forward is built and tested in the PoC; the event bus, database standby and crop object store are design.



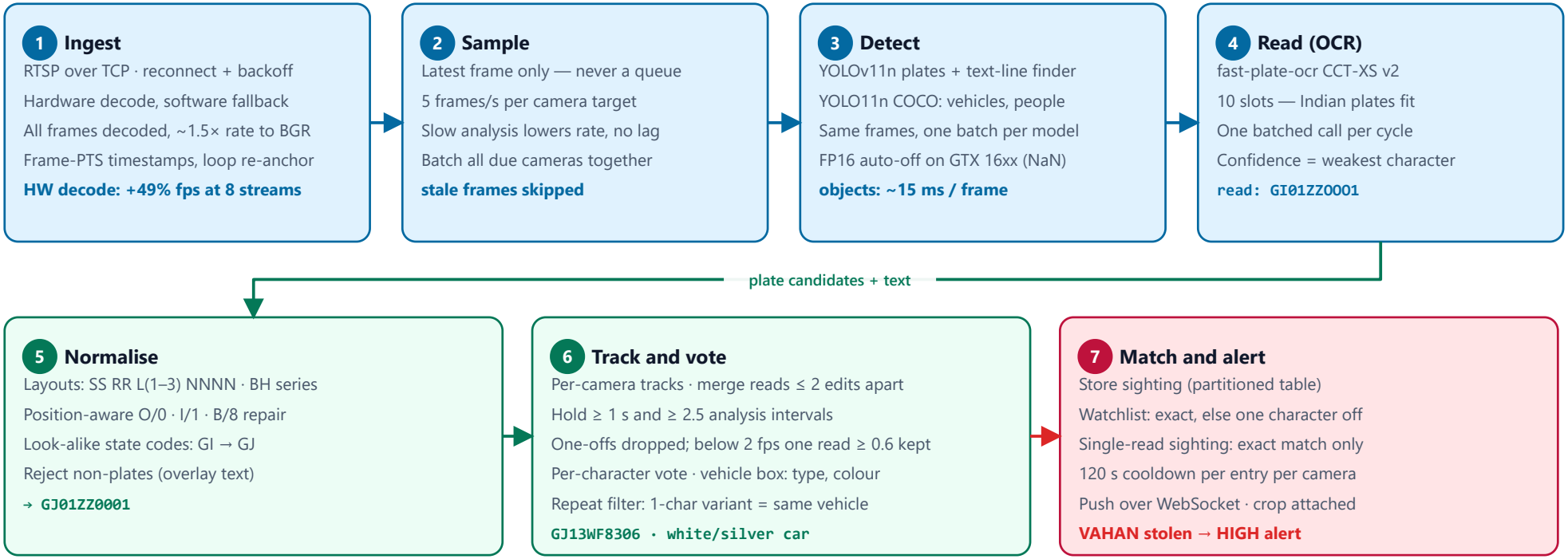
Integration workflow: onboarding a camera by stream URL

The camera sees one session; credentials never leave the server; a department user can onboard only their own department's cameras. Measured: 1.2–2.0 s end to end on an idle stack (17 Sep).



Analytics workflow: from RTSP frame to watchlist alert

Hardware decode where available; plate detection and OCR, and vehicle and person detection, on the same frames; a per-character vote across each track's reads; a repeat filter; matching and alerting. Laptop measurements, 24 Sep.



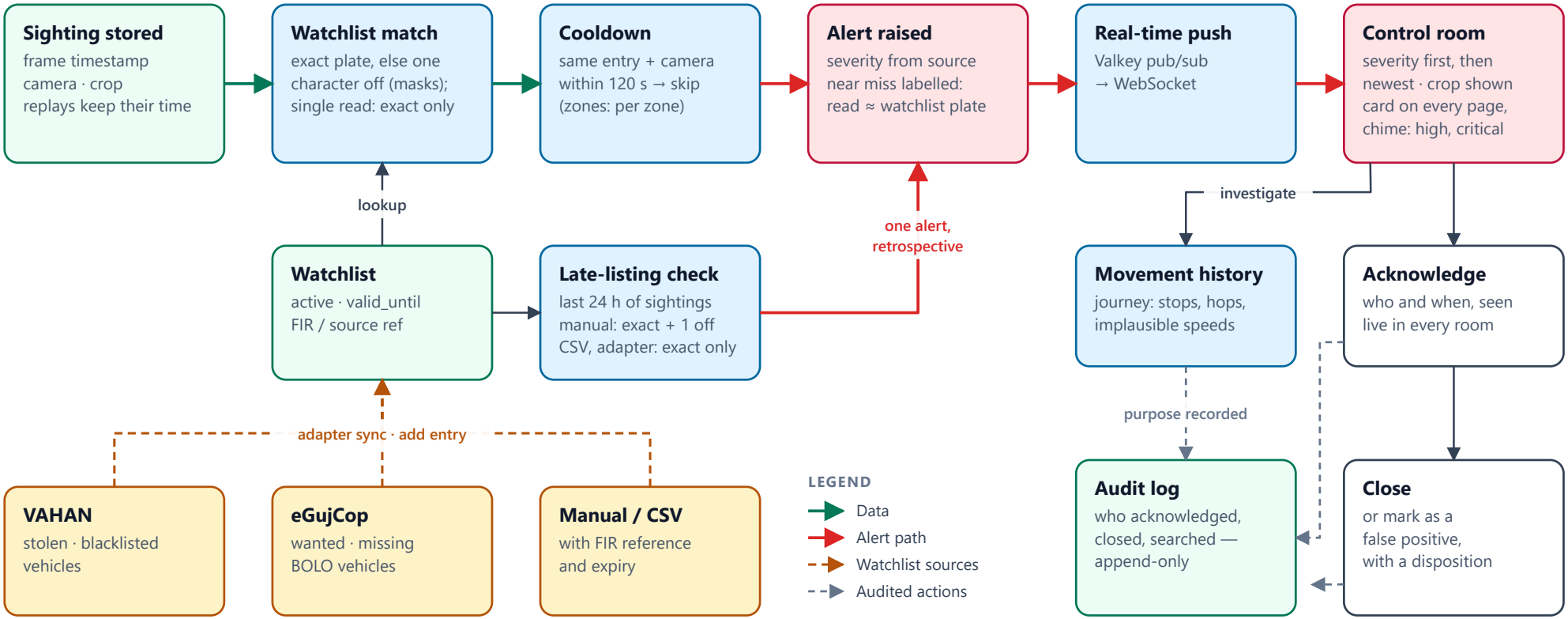
Measured 24 Sep on a GTX 1650 Ti laptop, 4 cameras, 15 min: over 1,400 sightings, 31 watchlist alerts; first read → alert row median 1.06 s, p95 1.94 s.

Hardware decode (D3D11, the laptop's Intel iGPU, not NVDEC): +49% analysed fps per stream at 8 and 12 streams. Keeping frames on the GPU is production design.

API unreachable: batches spool to disk and replay in order with their frame times; hits on a cached watchlist are logged at the edge, alerted on replay.

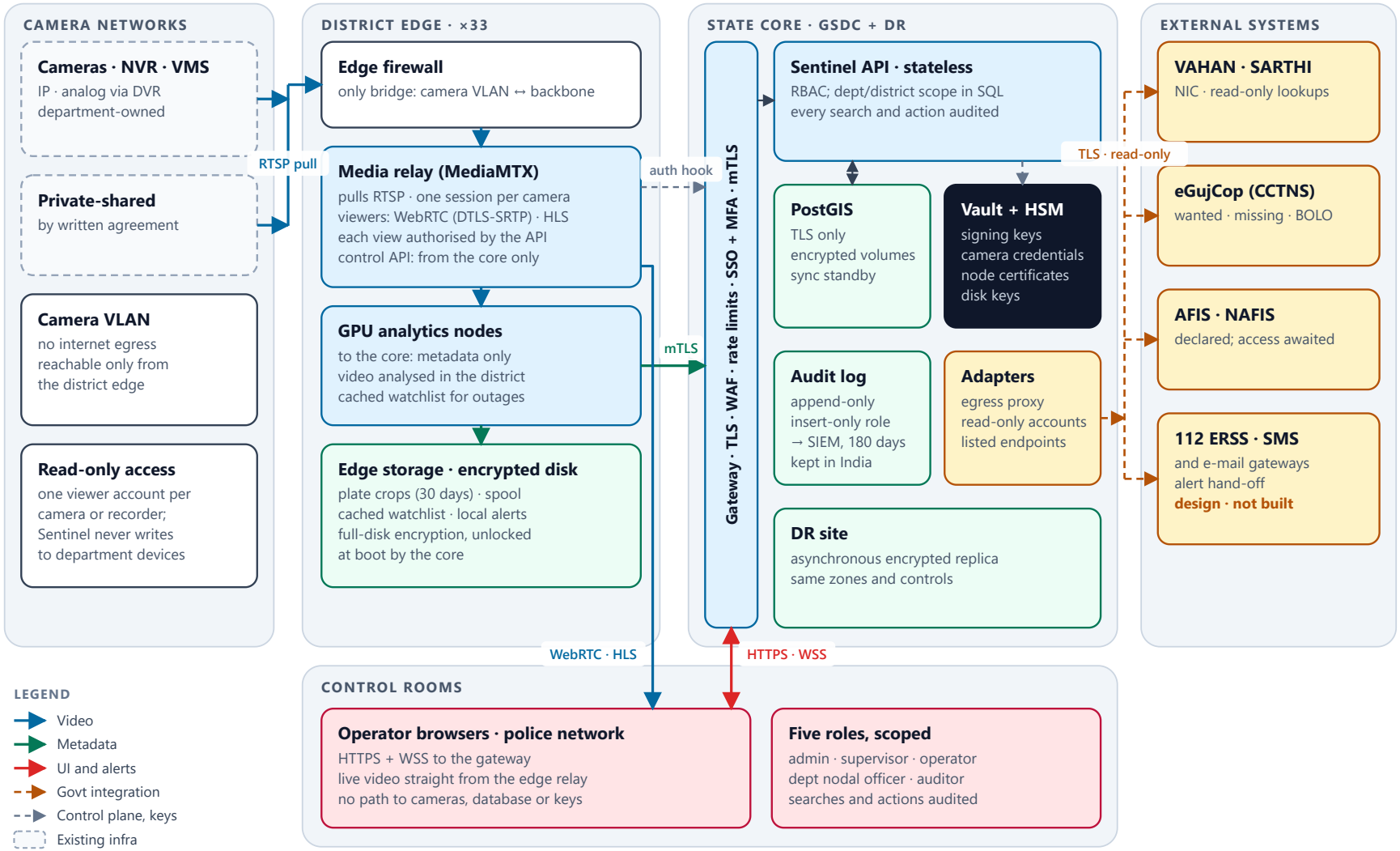
Alert workflow: from sighting to closed alert

Open alerts in severity order; near misses labelled; a plate added to the watchlist is checked against the last 24 hours. Every operator action is audited.



Security zones (target design)

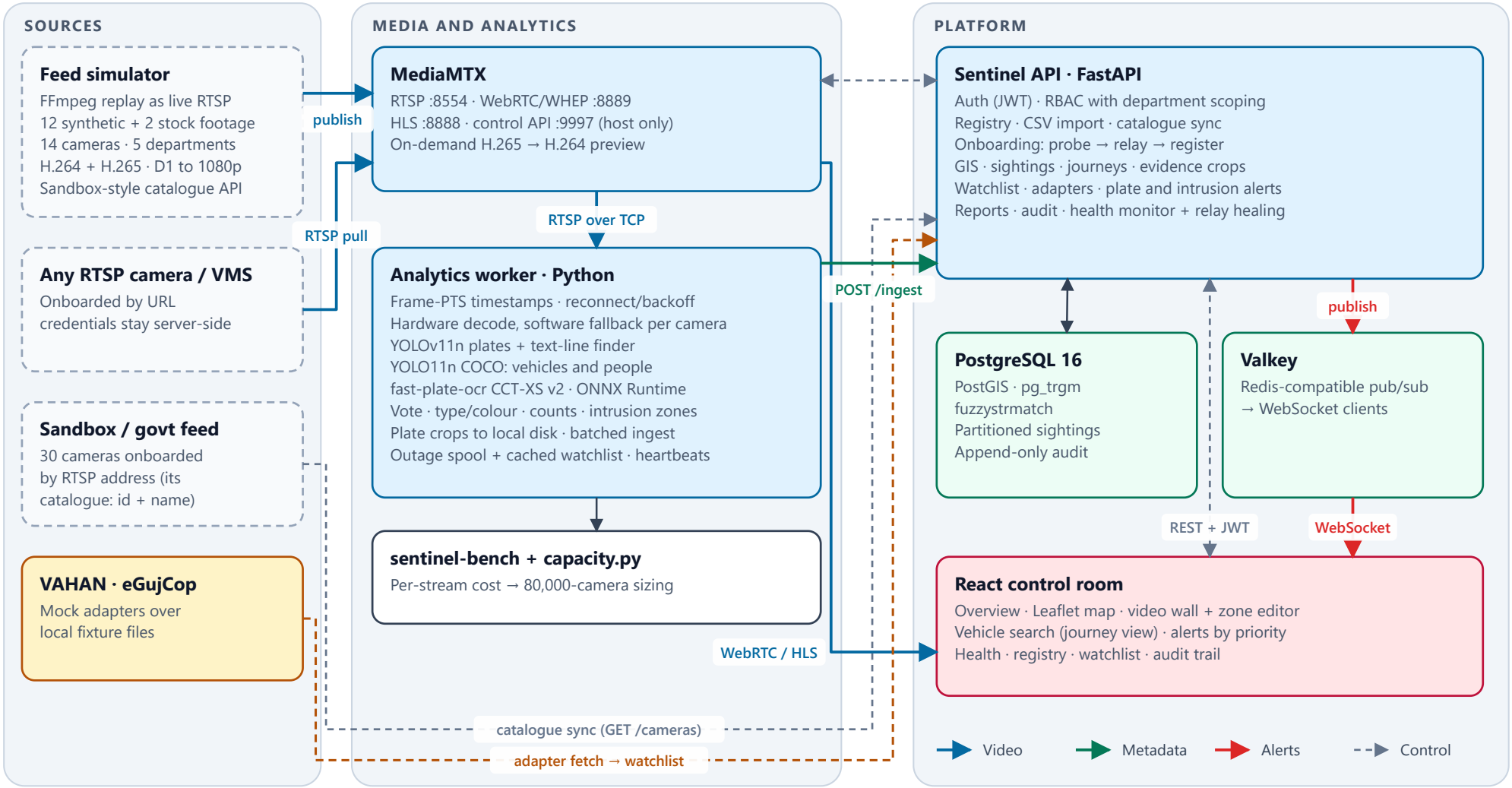
Each arrow is a permitted flow; video leaves the district only to authorised viewers. The PoC runs every zone on one laptop over plain HTTP: HLD 11.4 says what is built and what is not.



Target design. The PoC runs every zone on one laptop over plain HTTP, with one shared ingest key and one HS256 secret: HLD 11.4 says what is built and what is not.

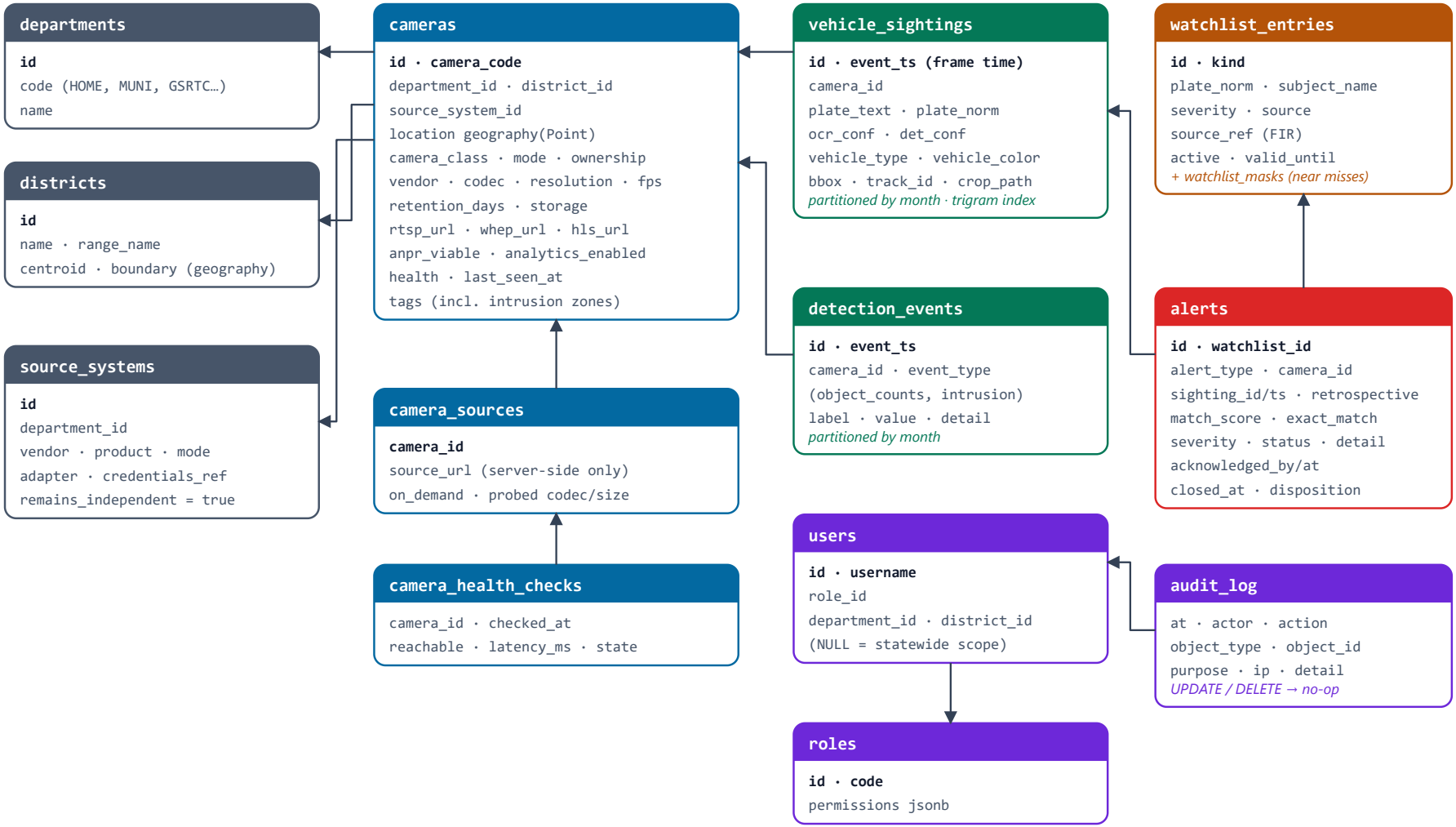
Proof of concept, as built

Every box is running code in the repository. VAHAN and eGujCop are mock adapters; two cameras are stock footage with synthetic plates.



Core data model

Video is not stored in the platform; sightings and counts are partitioned by time.



Video is not stored here: it stays on department recorders. The platform holds registry, metadata, evidence crops and the audit trail.

Statewide sizing

Derived in the Plan for Scale document from bench/capacity.py. Streams per node, tier shares and unit prices are ASSUMPTIONS until measured or quoted.

80,000 CAMERAS · TIERED BY WHAT EACH CAMERA'S POSITION ALLOWS

A · ANPR	B · Events	C · View and record
8,000 · 5 fps	16,000 · 1 fps keyframes	56,000 · relayed only while watched · no continuous compute

DISTRICT EDGE · 33 DISTRICTS · 248 GPU NODES



≈ 7.5 GPU nodes per district, co-located with the district control room

Relay + analytics + local spool + crop store · video stays inside the district network

Node count = tier A ÷ 60 streams + tier B ÷ 250 streams, + 25% headroom (assumptions to be measured)

metadata 2.7 Mbps

STATE CORE

GSDC primary

API cluster behind load balancer
PostGIS primary + synchronous standby
Kafka · watchlist matcher · audit
Government DB adapters

replication

DR site

Asynchronous replica · RPO minutes
RTO target under 4 hours

56,000	2.7 Mbps	2.6 TB	11.5 TB	≈ ₹39.7 cr
inference frames/s	metadata backhaul	sightings, 90 days	plate crops, 30 days	capex, assumed prices
statewide, tiers A + B	average, all districts	48 M rows/day, partitioned	kept at the edge	248 nodes × ₹15 lakh + core